```
$ hadoop jar /usr/share/java/hadoop/hadoop-mapreduce-
examples.jar teragen 10000000 gendata
$ hadoop jar /usr/share/java/hadoop/hadoop-mapreduce-
examples.jar terasort gendata sortdata
$ hadoop jar /usr/share/java/hadoop/hadoop-mapreduce-
examples.jar teravalidate sortdata reportdata
$ hadoop fs -rm -r gendata sortdata reportdata
```

Stop the Hadoop services before creating and working with multiple data nodes, and clean up the data directories:

```
$ sudo systemctl stop hadoop-namenode hadoop-datanode \
      hadoop-nodemanager hadoop-resourcemanager
$ sudo rm -rf /var/cache/hadoop-hdfs/hdfs/dfs/*
```

Figure 1: OpenStack-Hadoop

Testing with multiple nodes

The following steps simplify creation of multiple instances:

- Generate *ssh* keys for password-less log in from any node to any other node.

```
$ ssh-keygen
$ cat .ssh/id_rsa.pub >> .ssh/authorized_keys
```

- In */etc/ssh/ssh_config*, add the following to ensure that *ssh* does not prompt for authenticating a new host the first time you try to log in.

```
  StrictHostKeyChecking no
```

- In */etc/hosts*, add entries for slave nodes yet to be created:

```
192.168.32.2  h-mstr  h-mstr.novalocal
192.168.32.3  h-slv1  h-slv1.novalocal
192.168.32.4  h-slv2  h-slv2.novalocal
```

Now, modify the configuration files located in */etc/hadoop*.

- Edit *core-site.xml* and modify the value of *fs.default.name* by replacing localhost by *h-mstr:*

```
<property>
   <name>fs.default.name</name>
      <value>hdfs://h-mstr:8020</value>
 </property>
```

- Edit *mapred-site.xml* and modify the value of *mapred.job.tracker* by replacing localhost by *h-mstr:*

```
<property>
    <name>mapred.job.tracker</name>
    <value>h-mstr:8021</value>
  </property>
```

- Delete the following lines from *hdfs-site.xml:*

```
  <!-- Immediately exit safemode as soon as one DataNode
checks in.
    On a multi-node cluster, these configurations must be
removed. -->
    <property>
    <name>dfs.safemode.extension</name>
    <value>0</value>
  </property>
  <property>
     <name>dfs.safemode.min.datanodes</name>
     <value>1</value>
  </property>
```

- Edit or create, if needed, slaves with the host names of the data nodes:

```
[fedora@h-mstr hadoop]$ cat slaves
h-slv1
h-slv2
```

- Add the following lines to *yarn-site.xml* so that multiple node managers can be run:

```
  <property>
    <name>yarn.resourcemanager.hostname</name>
    <value>h-mstr</value>
  </property>
```

Now, create a snapshot, Hadoop-Base. Its creation will take time. It may not give you an indication of an error if it runs out of disk space!